Increasing electricity exports through delivery-informed allocation in distribution networks

Junjie Huang[a], Tao Yu[a], Zhenning Pan[a,*], Yufeng Wu[a], Wencong Xiao[a], Rankai Zhu[a] and Jie Huang[b]

[a]*School of Electrical Engineering, South China University of Technology, and Guangdong Provincial Key Laboratory of Intelligent Measurement and Advanced Metering of Power Grid, Guangzhou, 510640, China*

[b]*NARI Technology Co., Ltd., Nanjing, 211106, China*

ARTICLE INFO

Keywords:
Distributed generation
Electricity exports
Distribution networks
Capacity allocation
Counterfactual feedback

Abstract

Export limits assigned before generation is known can remain unused while other generators are curtailed. When limits remain fixed within an operating interval, the operator must decide whether an observed shortfall should reduce the generator's next request limit. Such a restriction can release access elsewhere but also suppress export as output recovers. We propose externality-conditioned counterfactual feedback (ECF) to guide this decision without changing the allocation rule. ECF recalculates the completed interval with lower requests from generators that left awards unused. Available-output estimates quantify the delivery changes elsewhere and in total, while nonlinear alternating-current power flow checks feasibility. A feasible replay that satisfies the access and delivery-gain conditions triggers gradual request-limit reductions; otherwise, limits recover toward the connection bounds. Controlled comparisons use a common output model for request choice and subsequent availability. At zero shortfall cost and update weight 0.5, ECF retains 79.4% of the additional delivery obtained without feedback by participants submitting larger requests, relative to forecast-based requests. Mean total delivery rises above that reference. ECF also delivers more than proportional reductions of the same total request in 95.6% of 90 matched trajectories across five allocation rules. An 80-day Australian public-data simulation provides a further comparison, with 1.39 MWh more modeled export than no feedback. Immediately replacing ceilings with low-output targets reverses the benefit in the tested cases. These results support selecting request-limit updates by their delivery consequences and assessing their effects through both low-output and recovery periods.

1. Introduction

1.1. Operating context and motivation

Homes and businesses with rooftop photovoltaics (PV) can meet part of their own electricity demand, but exporting the surplus requires access to a shared distribution network. Voltage and equipment limits constrain that access. Reinforcement and flexible power-flow control can expand the operating range [1]; allocation determines how generators share access within that range. China's distributed-PV management framework also places connection capacity and operational control alongside project development [2]. Making effective use of existing export capacity therefore complements new generation and network reinforcement.

Flexible export arrangements make this allocation an ongoing operating task. South Australia's flexible-export programme offers an alternative to restrictive fixed limits [3], and California's limited generation profiles permit export schedules aligned with time-varying hosting capacity [4]. The European Union also provides for flexible connection agreements where network capacity is limited [5]. These arrangements use different procedures to adjust permitted export to network conditions. Operating envelopes specify power limits at each connection, within which owners or aggregators schedule their resources [6–8]. For arrangements that retain an allocation throughout an operating interval, an additional operating question arises after delivery: whether an unused allocation should change access in the next interval.

*Corresponding author.

Email addresses: junjiehuang1@outlook.com (J. Huang); taoyu1@scut.edu.cn (T. Yu); scutpanzn@163.com (Z. Pan); wuyuffeng@163.com (Y. Wu); 202311083113@mail.scut.edu.cn (W. Xiao); ep0129rkzhu@mail.scut.edu.cn (R. Zhu); huangjie1@sgepri.sgcc.com.cn (J. Huang)

We study request-based arrangements in which participants seek access before their output is known. A related information sequence appears in Project Converge's shaped-operating-envelope design: aggregator bids and customer contribution plans precede the calculation of export bounds [9]. We call the allocated export-power limit an *export award*; access and authorization refer to this allocated permission. A shortfall is the unused part of the award, rather than an error relative to the generation forecast. The award remains fixed until the interval ends, so its unused portion is not reassigned during that interval. A request above the point forecast can allow more export when output is high but leave part of the award unused when output is low. The larger award can also reduce the allocation to another generator. The requester's additional export may then be smaller than the export lost elsewhere. This creates a recurring choice for the operator. Reducing a participant's request limit after every shortfall can constrain its export when output recovers. Leaving the limit unchanged can continue to restrict other generators with available output. The unused amount alone does not tell the operator which response will improve delivery. The practical need is a criterion for carrying a completed shortfall into a future restriction, with a way to restore access as generation changes.

Figure 1 shows why the unused amount cannot make this distinction. Connections A and B share a binding limit, and B leaves the same amount unused in (b) and (c). Reallocation increases delivery only in (c), where A has additional available output. Neither B's shortfall nor A's delivery at its limit reveals that opportunity. In a network, electrical location also matters because additional export affects voltage and equipment loading differently across connections. Estimating how much additional electricity could have been exported therefore requires available-output estimates at the affected connections and a network-feasible alternative allocation.

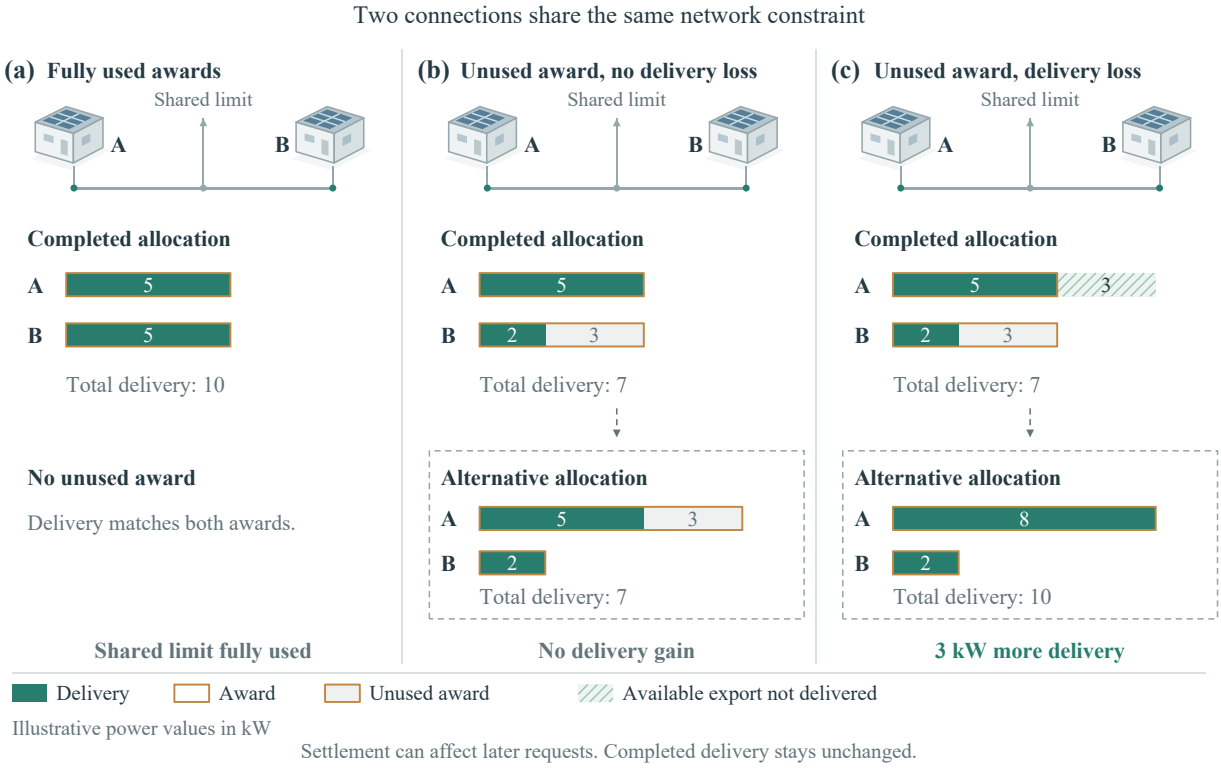


Figure 1: When unused export allocations reduce delivery elsewhere. Two connections share a binding network constraint. (a) Both awards are fully used. (b) B leaves part of its award unused, but A has no additional available export. An alternative allocation increases A's award without increasing delivery. (c) B leaves the same portion unused while A has available export that its award prevents it from delivering. The alternative allocation increases total delivery. Teal fills show delivery, amber outlines show awards, and hatching shows available export not delivered. Numbers within each segment give its power in kW. Dashed frames show calculations for the completed interval. Settlement can affect later requests but cannot restore that interval's lost delivery. The illustrative values use a shared 10 kW export limit and initial awards of 5 kW each; they are not case-study results.

1.2. Related work and unresolved operating question

Research on export access addresses both network feasibility and the consequences of sharing limited capacity. Andoni et al. [10] show how curtailment rules affect generation investment and profitability. Gorrasi et al. [11] compare static limits, dynamic operating envelopes and distribution locational marginal prices, finding that the value of a more complex market design depends on network structure and customer preferences. These studies make the allocation rule an engineering and economic choice, rather than treating feasible export capacity as a single interchangeable quantity.

Robust operating envelopes account for uncertain injections when defining network-feasible bounds [12, 13], while network-secure bidding combines those bounds with adjustments to uncertain output [14]. Allocation objectives can favor total access, proportional sharing or equality [15–18]. Market mechanisms coordinate bids and prices with network constraints and participant incentives [19–21]. Project EDGE's comparison of forecast update frequencies further shows why the timing and accuracy of information matter for access calculation [22]. These approaches improve forecasts, feasible limits or allocation objectives before delivery. A shortfall observed later poses a different decision. The operator must determine whether changing future requests is likely to release useful access, rather than simply respond to a temporary reduction in generation. Both situations can produce the same unused award.

History-dependent allocation and access exchange already provide ways to redistribute access. Fair-over-time coordination spreads curtailment across participants and intervals [23]. Gupta and Molzahn [24] use previous curtailment to weight next-day topology reconfiguration, improving fairness while limiting additional curtailment. A recent preprint on stateful pricing and allocation uses historical delivery relative to available export to adjust subsequent priorities [25]. SecuLEx instead permits network-checked exchange of issued limits according to customer needs [26]. These approaches already use history or changing needs to redistribute access. The decision examined here is whether an observed shortfall warrants restricting subsequent requests while retaining the allocation rule. A shortfall measures unused permission; establishing a delivery opportunity also requires available generation elsewhere and a feasible redistribution. We calculate that alternative for the completed interval, accounting for both additional delivery elsewhere and any delivery removed from the unused-award group. This supplies a criterion for selecting updates that complements the use of history in allocation.

Settlement addresses the incentive to request access. A charge based only on B's shortfall gives the same signal in Figure 1(b) and (c), although only one contains recoverable delivery. A charge informed by system-wide delivery effects would also need to estimate the export lost elsewhere. Faster reassignment can address a mismatch before the interval ends. Where awards remain fixed, measurement-based coordination instead allows completed records to inform later allocations [27, 28]. Two comparisons are therefore needed. Recalculating the completed interval tests whether reallocation could have increased export. Following subsequent intervals tests whether lower request limits actually increase export as generation changes. A benefit in the first comparison does not guarantee a benefit in the second, because a later restriction can coincide with recovering output.

1.3. Study objective and contributions

We develop ECF to select when completed delivery should lower future request limits. Its decision criterion is a network-feasible alternative allocation that would have increased delivery both outside the unused-award group and in total. The resulting update is gradual, and limits recover toward the connection bounds when the criterion is not met. This combines a test of the completed shortfall with control of the restriction carried into later intervals. It can be applied while retaining the allocation objective and network limits; completed awards are left intact.

The analysis links request choice, allocated access and delivery through one conditional output distribution. A request–access–delivery externality requires both the changed group's payoff and unused award to increase while total delivery falls. This model provides an explicit basis for the controlled requests and their subsequent output. The operator's ECF calculation instead uses completed records and available-output estimates.

The comparisons distinguish the choice of participants, the choice of update intervals and the update strength. Controlled equal-total reductions compare targeting across five allocation rules within the prescribed allocation domain. In the Australian public-data simulation, updating after every shortfall and comparably frequent random updates test whether the replay criterion helps select useful intervals. Gain comparisons then

expose the delivery lost when restrictions persist into generation recovery. Together, these results evaluate the decision to impose a later restriction and its consequences, rather than judging feedback by unused capacity alone. Section 2 presents the model and feedback method; Section 3 specifies the experiments and comparisons. Sections 4–6 present the results, operating implications and conclusions. Supplementary information gives the full derivations and additional sensitivity analyses.

2. Model and feedback method

Figure 2 connects the request, allocation and delivery quantities defined below to the subsequent ceiling update.

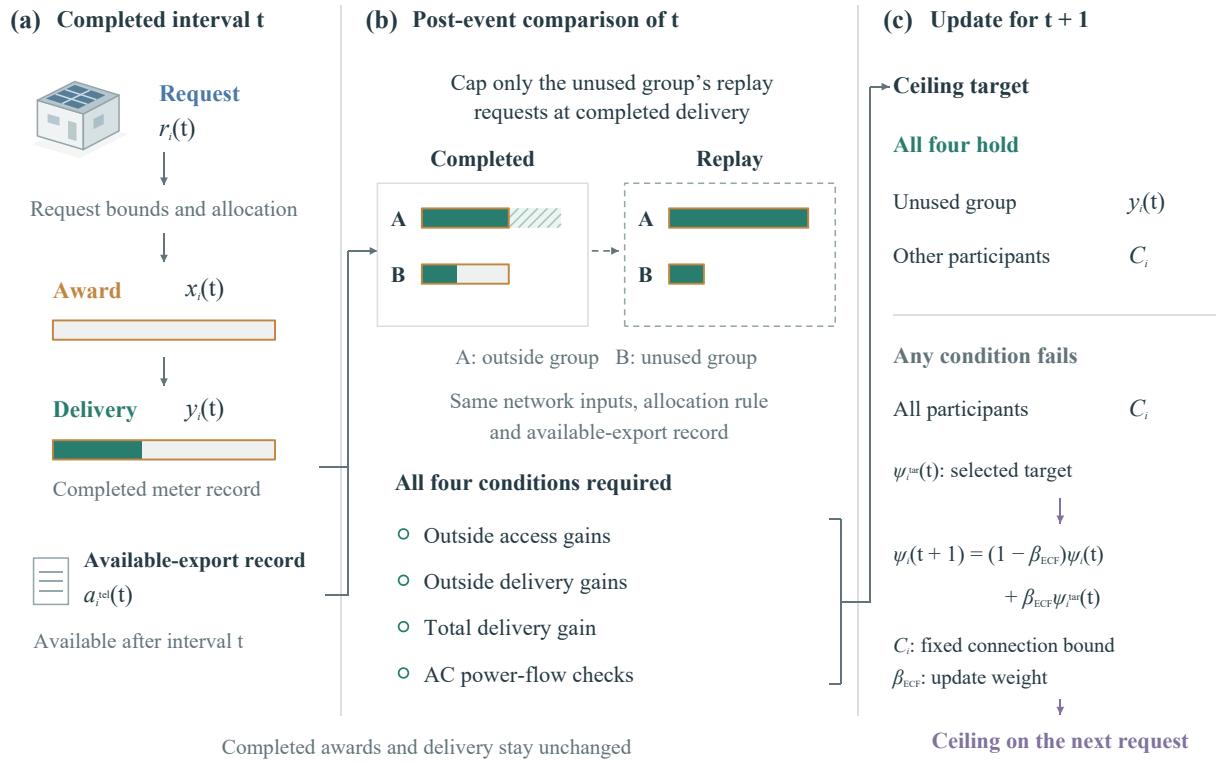


Figure 2: Using completed delivery to update later request ceilings. (a) Connection and feedback bounds limit requests before the existing rule allocates access. (b) The unused-award group's requests are reduced to completed delivery, and allocation is repeated under the same exogenous network conditions. Available-export estimates determine the resulting delivery. An update requires positive changes in access and delivery outside the group, summed over the connections that gain, a net increase in total delivery and AC feasibility. (c) When these conditions hold, the group's next request ceilings move toward its completed delivery; other ceilings move toward the fixed connection limits. If the conditions do not hold, all ceilings move toward those limits. Completed awards and delivery stay unchanged.

2.1. Requests, awards and available export

Let $\mathcal{N}$ be the participating export connections and $\mathcal{G} = (\mathcal{B}, \mathcal{L})$ the distribution network, with buses $\mathcal{B}$ and branches $\mathcal{L}$. In interval t, participant i requests $r_i(t)$ before its available export $a_i(t)$ is known. Available export is the power that the participant could supply at its connection before applying the allocated export limit. Its fixed connection export limit is C_i. Feedback changes a request ceiling $\psi_i(t) \in [0, C_i]$, initially C_i, so the admitted request is

$$r_i^{\mathrm{adm}}(t) = \min\{r_i(t), C_i, \psi_i(t)\}. \tag{1}$$

The allocation rule grants export awards $\mathbf{x}(t)$ from these admitted requests. Completed delivery is $y_i(t) = \min\{x_i(t), a_i(t)\}$, with $0 \le y_i(t) \le x_i(t) \le r_i^{\mathrm{adm}}(t) \le \psi_i(t) \le C_i$. Thus $x_i - y_i$ is the unused award. An award

is permitted power, and delivery is power actually exported. Total delivery sums exports at the participant connections; feeder losses and upstream net exchange are accounted for separately.

Awards remain fixed within an interval, with no redistribution of access. Feedback modifies only the admitted-request input in a later interval, not the rule or network limits. These variables are interval-average powers. Multiplication by interval duration and summation give the reported energy totals. Table S1 collects the quantities.

For rule $\mathcal{R}$, the allocation map is $\mathbf{x}(t) = \Pi_{\mathcal{R}}(\mathbf{r}^{\mathrm{adm}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \boldsymbol{\eta}_{\mathcal{R}})$. Here $\mathcal{M}$ denotes the network model, and $\mathcal{F}_{\mathcal{M}}$ its equations and limits under exogenous data $\mathbf{z}(t)$. The vector $\boldsymbol{\eta}_{\mathcal{R}}$ contains rule parameters, held fixed in the comparisons.

The coordinator applies the allocation rule and updates request ceilings. It receives requests before allocation and delivery records after the interval. Recalculating delivery under a different award also requires an estimate of the export power that was available in that interval, denoted $a_i^{\mathrm{tel}}(t)$ and called the available-export record. Participants' private output signals and conditional distributions are not inputs to the coordinator. The following assumptions specify the record and the conditions under which it can support that comparison.

Assumption 1 (Post-event available-export information)**.** *The comparison uses an available-export record a_i^{tel} for the completed interval. The record must describe available export at the same connection and interval, including output that the issued award prevented from being delivered. In the controlled study the record is exact simulated availability; in the Australian study the primary record is the profile-derived proxy a_i^{prof}. Record-error tests vary the replay input a_i^{tel}, while unperturbed a_i^{prof} continues to generate modeled delivery. Neither record changes an award already issued.*

Assumption 2 (Same-state replay)**.** *The replay changes only admitted requests when recalculating the completed interval. Topology, exogenous network data, participants, rule parameters and underlying available export remain fixed. Awards and power flows are recalculated. If a different award would change availability through inverter controls, storage, demand or portfolio dispatch, those responses must instead enter the replay model.*

2.2. Request choice and delivery displacement

The request model explains why an individually useful request can reduce total delivery. It is used to construct the controlled comparisons; the coordinator does not estimate this private model to run ECF. Participant i observes a private signal s_i, forms a point forecast a_i^{fcst}, and assigns conditional cumulative distribution function (CDF) $F_i(\cdot \mid s_i)$ to its available export. Let $r_i^{\mathrm{ref}} = a_i^{\mathrm{fcst}}$ be its point-forecast request and $r_i^{\mathrm{chg}} = \gamma_i r_i^{\mathrm{ref}}$, $\gamma_i \geq 1$, a larger request. Holding other requests and exogenous network data fixed, the participant values own delivery and faces a modeled shortfall cost $\vartheta \geq 0$ per unit of unused award, normalized by the value of delivered export:

$$U_i(\gamma_i; \vartheta) = \mathbb{E}\left[\min\{x_i(\gamma_i), a_i\} - \vartheta[x_i(\gamma_i) - a_i]^+ \mid s_i\right]. \tag{2}$$

Here $[v]^+ = \max\{v, 0\}$ is the positive part. When several multipliers give the largest expected payoff, we select the smallest on the prespecified grid Γ. The same conditional distribution determines the optimal request and the subsequent available export. The request choice therefore reflects the uncertainty in the generation that may use the allocation. A static changed-request case is classified as a request–access–delivery externality if the changed group's expected payoff and unused award both rise while aggregate expected delivery falls beyond the numerical tolerance. A productive high request raises the group's payoff without such a delivery loss. Dynamic cases use realized counterparts. This economic classification is distinct from ECF's operational replay test.

Where the allocation map is differentiable and the available-export CDFs are continuous at the current awards, write $J_{ji} = \partial x_j / \partial r_i$ and $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} \mid s_j]$ for expected total delivery. Here $F_j(x_j)$ abbreviates the conditional cumulative distribution $F_j(x_j \mid s_j)$. Then

$$\begin{aligned} \frac{\partial U_i}{\partial r_i} &= J_{ii}[1 - (1 + \vartheta)F_i(x_i)], \\ \frac{\partial \bar{Y}}{\partial r_i} &= \sum_j J_{ji}[1 - F_j(x_j)]. \end{aligned} \tag{3}$$

The factor $1 - F_j(x_j)$ is the probability that connection j can use additional allocated power. The allocation derivative J_{ji} determines how much of that power it gains or loses when i requests more. The private-payoff and total-delivery derivatives can therefore have opposite signs. If the requester gains expected delivery, total delivery falls when the net expected loss at other connections exceeds that gain. Both the sizes of the allocation changes and their probabilities of use matter. This is an allocation effect; the delivery sum is measured at the connections and does not subtract feeder losses. Network constraints influence it through the allocation map. The equations do not identify which constraint is binding in a particular numerical case.

The numerical study uses discrete output values, and a request change can switch the binding constraints. We therefore classify the reported cases using finite differences between matched requests. When additional outside-delivery loss and additional unused award are both positive, their ratio measures the loss elsewhere per additional unit left unused. This is the quantity plotted in Figure 3(d). Its finite-change definition and relation to the local marginal shortfall cost are in supplementary section S2; that section also defines the capacity-normalized Jain index.

2.3. Network-constrained allocation

Five rules use the same admitted requests and feasible allocation domain in the controlled study (Table S2). Weighted proportional (WP) scales all requests by one feasible factor. Equal reduction (ER) subtracts a common amount; equal access (EA) applies a common cap. Maximum export (ME) maximizes total allocated export power. Fairness-aware allocation (FA) balances total allocated power against inequality in the fraction of each request served. Their one-parameter formulas, conic objectives, deterministic tie-breaks and sensitivity derivations are specified in supplementary sections S5–S8. Section S4 derives the matched request-reduction comparison. The controlled implementation uses an affine network approximation with second-order-cone constraints on two-ended branch loading, calibrated approximation margins and a prescribed bound on normalized allocation size. That domain bound limits the allocations considered by the approximation; it is separate from equipment ratings. Nonlinear AC power flow separately verifies the resulting states. The feeder-specific models and physical checks are specified in Section 3.2 and Section 3.3.

2.4. Completed-interval replay and feedback

Replay and request-ceiling update. ECF tests an alternative allocation for the completed interval before updating future request ceilings. With unused-award tolerance $\epsilon_x > 0$, let $\mathcal{N}_{\mathrm{ua}}(t) = \{i : x_i(t) - y_i(t) > \epsilon_x\}$. After the interval ends, ECF caps only this group's admitted requests at completed delivery:

$$r_i^{\mathrm{rpl}}(t) = \begin{cases} \min\{r_i^{\mathrm{adm}}(t), y_i(t)\}, & i \in \mathcal{N}_{\mathrm{ua}}(t), \\ r_i^{\mathrm{adm}}(t), & i \notin \mathcal{N}_{\mathrm{ua}}(t). \end{cases} \tag{4}$$

The same allocation rule and network data give recalculated awards $\mathbf{x}^{\mathrm{rpl}}$. Using the available-export record, their corresponding delivery is

$$y_i^{\mathrm{rpl}} = \min\{x_i^{\mathrm{rpl}}, a_i^{\mathrm{tel}}\}.$$

The replay must pass the applicable nonlinear AC checks and satisfy three gain conditions. The two outside-group quantities sum positive changes in awards and delivery over connections that gain; they are not net changes across all outsiders. The third quantity is the net change in total delivery, including any group member's loss under a replay award below its completed delivery. Each increase must exceed τ_{rpl} (supplementary section S3). Completed awards and delivery remain unchanged.

If all four conditions hold, set $g(t) = 1$ and use completed delivery as the unused-award group's ceiling target. All other targets are the fixed connection limits. The indicator is zero if any condition fails. The update is

$$\psi_i^{\mathrm{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\mathrm{ua}}(t), \\ C_i, & \text{otherwise}, \end{cases}$$
$$\psi_i(t+1) = (1 - \beta_{\mathrm{ECF}})\psi_i(t) + \beta_{\mathrm{ECF}}\psi_i^{\mathrm{tar}}(t), \qquad 0 \le \beta_{\mathrm{ECF}} \le 1. \tag{5}$$

The replay selects whether to update; the gain controls how much of that update carries into the next interval. For positive gain and $g(t) = 1$, ceilings decrease for the unused-award group; all others move toward connection limits. If $g(t) = 0$ persists, the gap to each connection limit decreases geometrically. Admitted requests cannot bypass a ceiling. The test concerns the group's joint reduction, not each member's contribution. A positive completed-interval comparison does not predict the next interval's availability. Subsequent delivery must therefore be evaluated along the feedback trajectory, including output recovery.

Available-export record and uncertainty. With exact completed delivery and a validated record-error bound $|a_i^{\mathrm{tel}} - a_i| \le \delta_i$, the conservative replay replaces the record by

$$\max\{y_i, a_i^{\mathrm{tel}} - \delta_i\}$$

as a lower bound on available export. It gives a lower bound on the delivery increase under exact availability; nonlinear AC checks remain necessary. With exact records and a common threshold, a total-delivery increase implies the two outside-group increases. All three quantities are retained to distinguish allocation from delivery when records are perturbed. Supplementary section S3 gives both derivations.

Matched reduction benchmark and delivery accounting. The equal-total control matches ECF's total request reduction in each interval but spreads it proportionally across connections. If $r_i^{\mathrm{cap}} = \min\{r_i, C_i\}$ and $\Delta R_{\mathrm{ECF}} = \sum_i (r_i^{\mathrm{cap}} - r_i^{\mathrm{adm,ECF}})$, the control admits

$$r_i^{\mathrm{adm,uni}} = r_i^{\mathrm{cap}} \left(1 - \frac{\Delta R_{\mathrm{ECF}}}{\sum_j r_j^{\mathrm{cap}}} \right) \tag{6}$$

when the denominator is positive, and retains the capacity-limited request otherwise. Thus the controls reduce total requests by the same amount but reduce different participants' requests. Equal request reduction need not give equal allocation or delivery. With total authorization $X = \sum_i x_i$, total delivery $Y = \sum_i y_i$ and unused authorization $H = \sum_i (x_i - y_i)$, the accounting identity $Y = X - H$ yields

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = (X^{\mathrm{ECF}} - X^{\mathrm{uni}}) + (H^{\mathrm{uni}} - H^{\mathrm{ECF}}). \tag{7}$$

Summing over intervals, with each power multiplied by its interval duration, gives the corresponding energy balance. Either component can be positive or negative. Supplementary section S4 derives conditions under which redistribution increases delivery.

3. Simulation design and evaluation

The controlled study separates request incentives, allocation rules and feedback strength. Almost all of its replay attempts find a delivery gain, so it mainly tests where to reduce requests and how strongly to update. The Australian public-data study provides the main comparison of when to update, using household histories and an unbalanced network model. Within each comparison, controls share exogenous inputs and differ only in the specified request or feedback mechanism.

Table 1
How the comparison mechanisms differ. C denotes the controlled study and A the Australian simulation. Paired controls use the same network and available-export trajectory.

Mechanism	What changes	Question tested
No feedback (C, A)	Ceiling stays at the connection bound.	Baseline delivery.
Shortfall cost (C)	A modeled cost changes the initial request.	Effect of an incentive before allocation.
Mismatch-only feedback (C)	A participant's own delivery-to-award ratio updates its ceiling.	Effect of responding to shortfall alone.
Update without replay test (A)	ECF's formula applies after every unused award.	Effect of removing the test, including more frequent updates.
Frequency-calibrated random update (A)	The same formula updates randomly selected unused-award intervals.	Whether a similar update frequency explains ECF's gain.
Equal-total reduction (C, A)	ECF's total reduction is spread proportionally.	Value of where requests are reduced.
Previous-day persistence (A)	Request uses the previous-day output estimate alone.	Cost of a conservative request.
ECF (C, A)	A feasible replay showing increased delivery enables lower request ceilings.	Combined effect of selecting intervals and updating ceilings.

3.1. Comparison mechanisms

In the controlled study, two reference cases serve different purposes. The forecast-based reference uses point-forecast requests and no feedback. Comparing larger requests with this reference measures the effect of request choice. The no-feedback control instead retains those larger requests and disables the ceiling update. Comparing ECF with this control measures the effect of feedback. Table S9 places the controlled outcomes on both baselines.

The controls distinguish advance incentives from post-delivery updates. The shortfall-cost benchmark changes request choice and has no feedback state. Mismatch-only feedback smooths the participant's delivery-to-award ratio without replay. To remove replay without also changing the update formula, a separate control applies ECF's target and gain after every unused award.

To test whether a similar update frequency reproduces ECF's advantage, 12 fixed replay-free schedules select unused-award intervals with probability 483/3091, matching ECF's original update fraction among the 3,091 unused-award intervals, not among all 3,840 half-hours. This probability is calibrated retrospectively to make the update frequency comparable to ECF's. Each schedule evolves its own awards, delivery and ceilings, so update magnitudes and durations can also differ. Schedules are averaged within each date. Supplementary section S11.3 gives the selection procedure and all outcomes.

3.2. Controlled medium-voltage experiments

The controlled medium-voltage case uses the published radial 141-bus network [29], a 0.70 demand scale and 30 fixed connection points. Synthetic branch ratings, fixed before the reported comparisons, define the operating limits for this controlled experiment. Q0 denotes zero participant reactive power and Q95 denotes 0.95 power factor with reactive-power absorption. Each matched reference, changed-request, award, simulated-delivery and replay-delivery state is checked by nonlinear radial AC power flow at its fixed reactive-power setting. The check requires convergence, active- and reactive-power balance, $0.95 \le v_b \le 1.05$ pu, where v_b is bus-voltage magnitude, and

$$\frac{\max_{e\in\{\mathrm{fr,to}\}} |S_{\ell,e}|}{\bar{S}^{\mathrm{syn}}_{\ell}} \le 1 \quad \text{for every monitored branch } \ell. \tag{8}$$

The branch test uses apparent power in megavolt-amperes (MVA) at both ends, independently of rooted flow orientation; $\bar{S}^{\mathrm{syn}}_{\ell}$ is the assigned rating of branch ℓ. Supplementary section S8 gives the rating construction, calibration, and AC verification details.

The allocation approximation also imposes $\|\mathbf{D}_C^{-1}\mathbf{x}\|_2 \le 3.5$, where $\mathbf{D}_C = \mathrm{diag}(C_1, \dots, C_N)$ and $N = |\mathcal{N}|$. This is a specified allocation-domain bound, not a branch rating or a measured hosting-capacity limit. All five rules and their feedback controls use the same bound and calibrated approximation margins. The experiment tests reallocation within this joint domain. Section 4.3.1 reports how close the saved dynamic allocations lie to its boundaries; the separate AC check tests the feasibility of the selected states.

A five-outcome probability model for available export uses standardized values $(-2, -1, 0, 1, 2)$ with probabilities $(1, 4, 6, 4, 1)/16$ and scale σ relative to the participant's forecast. The participant's forecast determines both its request and the output values in this distribution. The static parameter grid spans 30 locations, five rules, Q0 and Q95, five uncertainty levels and nine shortfall-cost levels, giving 13,500 request-choice cases. The selected payoff-maximizing multiplier is held fixed within a 16-interval trajectory. Three temporal profiles preserve the occurrence count of each participant's output values but arrange them in different orders: clustered, intermittent and shock–recovery. Time offsets between participants also change which output values occur simultaneously across connections.

Changed requests come from one participant, five electrically nearby participants, or five dispersed participants. In the five-participant cases, each uses its own best-response multiplier and all five requests change simultaneously. The 270 base settings, each evaluated under six mechanisms, give 1,620 trajectories and 25,920 interval comparisons. Table S3 gives every factor, numerical tolerance and parameter level. The post-event record is exact simulated availability in this controlled study.

3.3. Australian public-data simulation

The public-data simulation combines a published unbalanced 11-kV feeder with 1,255 service connections and its low-voltage (LV) sections [30] with half-hour PV and consumption profiles from 300 Ausgrid solar homes [31]. These sources have no common customer identifier. The primary deterministic profile-to-service mapping is fixed before assessment, and separate placements test its sensitivity. The scenario represents increased PV adoption, with 418 exporting connections and PV systems scaled to an approximately 9.9-kWp mean [32]. Requests and awards represent this constructed scenario, not historical operating envelopes.

The day-to-day forecast increment is estimated from 2010–2011 data. Feeder settings use 2011–2012 data, and the reported comparisons use 2012–2013 profiles. A day is eligible only when mapped profiles are complete on that day and the previous day. Excluding missing and daylight-saving-transition records leaves 80 network days, or 3,840 half-hours per control. Initial exploratory runs also informed the operating setting; all reported 80-day comparisons use the resulting common setting. Supplementary sections S10.3–S10.4 give the construction sequence.

For mapped profile $\iota(i)$, source gross PV generation GG, general consumption GC and controlled load CL are first converted to interval-average MW. Applying the scenario multipliers gives

$$a_i^{\mathrm{prof}}(d,\tau) = \left[\lambda_{\mathrm{PV}}\mathrm{GG}_{\iota(i)}(d,\tau) - \lambda_L(\mathrm{GC}_{\iota(i)}(d,\tau) + \mathrm{CL}_{\iota(i)}(d,\tau))\right]^+. \tag{9}$$

Modeled delivery under control c is $y_i^c = \min\{x_i^c, a_i^{\mathrm{prof}}\}$; the unperturbed profile proxy also supplies the primary replay record.

Raw requests use the previous day's same-half-hour proxy plus a positive 90th-percentile increment from same-half-hour day-to-day changes. Contemporaneous household demand is fixed as an exogenous input to this retrospective allocation; load-forecast error is not modeled.

For AC delivery calculations, household consumption is first offset by local PV; the remaining net import minus delivered net export is the active-power load. The authorization screen instead sets participant net injection equal to the full award, without using realized PV to relax the constraint. Household reactive demand is retained in both screens. Supplementary section S10.4 gives the connection-level equations and mapping.

The primary feeder settings use a 0.30 load multiplier and a 1.07-pu allocation-stage upper-voltage limit, below the 1.10-pu physical limit. Together with the PV scaling, these settings construct an export-constrained operating scenario rather than reconstructing historical feeder operation. For each half-hour, the allocation algorithm starts from zero export and searches for a feasible common scaling factor for admitted requests, using nonlinear unbalanced AC power flow. An independent factor-grid check of all 19,200 primary allocation calls found no feasible sampled factor above the returned one; supplementary section S10.4 states the search resolution and its limits. Load-only, authorized-export and modeled-delivery states are screened separately

for convergence, customer voltage and native transformer ratings. Native line ampacities are unavailable, so the public-study screen excludes line thermal limits. The 2 W replay threshold suppresses numerical near-zero differences; record-error bounds are specified separately.

Each primary day begins with ceilings at C_i and compares no feedback, previous-day persistence, ECF at $\beta_{\mathrm{ECF}} = 0.5$, and equal-total reduction. The additional comparisons vary gain, the use of the replay test, cross-day memory, record perturbations and error bounds, profile placement, PV capacity, participant adaptation and published LV networks while retaining the documented fixed inputs for each matched comparison. Power at the feeder source, obtained from the same AC solutions, gives the net exchange with the upstream network.

3.4. Statistical analysis

The primary comparison unit in the Australian study is one paired network day. A day contains 48 half-hours on the same network and profile mapping; participants, half-hours and nested LV sections are not treated as independent network samples. Energy differences are computed within each day before averaging. We report ordinary paired-day percentile intervals and date-dependent wild-bootstrap intervals, which allow short-range correlation between nearby eligible dates. Both procedures use 10,000 replicates. For request adaptation, three seeds are averaged within each date. For the random-update comparison, 12 schedules are averaged within each date. Intervals are pointwise, without adjustment for multiple comparisons. Supplementary section S10.3 gives the calendar covariance, eligibility rules and numerical settings.

4. Results

Larger requests can protect one generator's export while causing a greater loss elsewhere. The controlled experiments show how targeted reductions recover part of that loss. Public-data simulations then distinguish the value of choosing update intervals from the benefit of updating less often. In both studies, stronger restriction can reverse the delivery gain. Table 1 identifies the alternative explanations tested by each control.

Unless a comparison explicitly names the forecast-based reference, feedback gains below are relative to no feedback or the named control. The reference-request comparison in Section 4.1 asks whether requesting more helps; the later comparisons ask whether feedback improves delivery from those requests.

4.1. Individual delivery gains can accompany a larger loss elsewhere

Larger requests increase the requester's expected delivery but reduce total delivery on average when there is no shortfall cost. Across 1,500 combinations of participant location, allocation rule, reactive-power setting and forecast uncertainty, the requester gains 5.2 kW relative to a point-forecast request. Other participants lose 18.8 kW. Total delivery falls by 13.5 kW, with all reported differences calculated before rounding. Both changes are calculated from the same allocation and probabilistic available-export model.

The loss occurs because the larger request changes the allocation to other connections, and those lost awards would have delivered more electricity than the requester gains. Equation (3) separates this allocation response from the probability of using each award.

Figure 3 traces the request choice as forecast uncertainty σ and normalized shortfall cost ϑ change. The cost parameter is the modeled charge per unit of unused allocated energy divided by the value per unit of delivered energy. The comparison covers the five allocation rules in Section 2.3 and both reactive-power settings. Each larger request is compared with the same participant's forecast-based request. Sections 2.2 and 3.2 specify the output model. Even at zero forecast uncertainty, a request above available output can compensate for rationing by an allocation rule. A multiplier above one therefore need not be explained solely by forecast error.

Individual cases differ from this mean. We classify a request–access–delivery externality when the requester's expected payoff and unused award rise while total delivery falls. A productive large request raises payoff without a total-delivery loss beyond the numerical tolerance. This classification concerns the effect of choosing a larger request; ECF instead tests whether a completed unused award could have supported delivery elsewhere. The definitions appear in Section 2.2 and supplementary section S2.

At $\vartheta = 0$, 71.5% of the 1,500 cases meet the externality definition, while 11.9% are productive large requests. The other 16.6% either retain the forecast-based request or meet neither classification. Thus request size and unused access alone cannot identify the delivery outcome. These shares pool all five uncertainty levels; Figure 3(c) fixes $\sigma = 0.20$ to show how the categories vary with shortfall cost.

A higher shortfall cost discourages requests likely to leave unused awards, but one charge cannot represent every loss to others. Panel (d) divides the additional outside-delivery loss by the additional unused award for a small increase in the request multiplier. Among steps that raise own delivery and unused awards while lowering total delivery, the ratio $\vartheta^{\mathrm{ext}}_{i,\Delta}$ has a pooled median of 2.29; its rule-specific medians differ substantially. Identical unused amounts can therefore accompany different losses elsewhere. Supplementary section S2 gives the full cost comparisons and ratio distribution. We next test whether feedback can reduce the outside loss while preserving useful export by the requester.

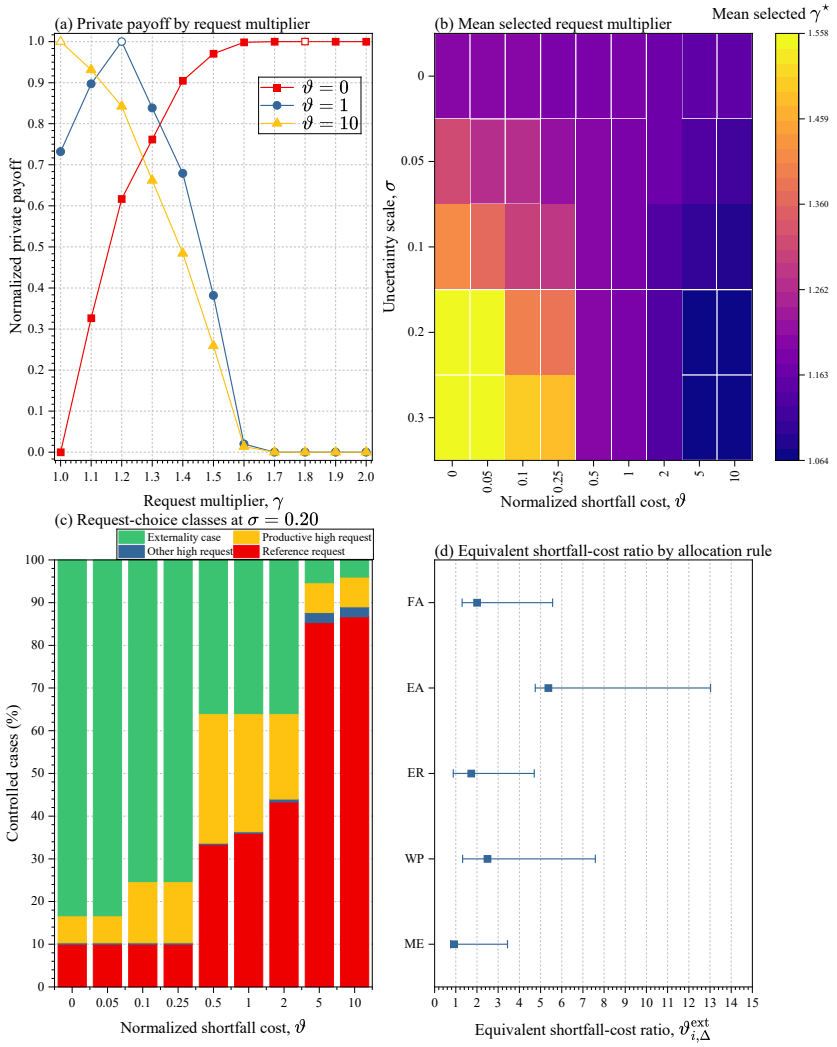


Figure 3: A larger request can protect own delivery or reduce total delivery. (a) Private-payoff curves for participant 21 under WP, Q0 (zero participant reactive power) and $\sigma = 0.20$, at three normalized shortfall-cost levels. Within each curve, payoff is rescaled as $(U - U_{\min})/(U_{\max} - U_{\min})$; a zero span is plotted as zero. (b) Mean selected request multiplier for each uncertainty and shortfall-cost setting. Means cover participants, rules and reactive-power settings. (c) Shares of four request classes at $\sigma = 0.20$ ($n = 300$). Reference requests use the point forecast. Productive high requests raise payoff without a total-delivery loss beyond the classification tolerance. Request–access–delivery externality cases raise payoff and unused awards while lowering total delivery. Other high requests meet neither definition. (d) Finite-difference cost ratio $\vartheta^{\mathrm{ext}}_{i,\Delta}$: expected delivery loss elsewhere per unit increase in expected unused award for a 0.1 request-multiplier step. Squares show rule-specific medians, and bars show interquartile ranges. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively.

4.2. Targeted request reductions preserve useful export access

At zero shortfall cost and the intermediate feedback gain, $\beta_{\mathrm{ECF}} = 0.5$, ECF delivers 24.8 kW more on average across the 16-interval trajectories than proportional reductions of the same total request. This is a 1.75% increase over the equal-total control's mean total delivery of 1416.4 kW. The following comparisons

separate the effects of the initial request choice, the placement of reductions and the strength of the update. Feedback controls share raw requests and network conditions; a shortfall cost instead changes the request choice before allocation. Figure S1 follows one trajectory as output falls and recovers.

4.2.1. A shortfall cost reduces displacement but also useful delivery

A shortfall cost changes requests before anyone receives an award. Raising ϑ from 0 to 1 reduces the mean multiplier from 1.51 to 1.14 and the share of intervals meeting the economic externality definition from 50.1% to 15.6%. But it also removes 45.6% of the larger-request group's delivery gain (Figure S2). At $\vartheta = 10$, none of these intervals meets the externality definition, while the group's gain is almost zero. The charge reduces export losses elsewhere but also removes allocated capacity that the group would use at high output.

4.2.2. Targeted feedback and its comparison baselines

At $\vartheta = 0$, ECF increases total delivery by 24.9 kW relative to no feedback. Its gain is 9.7 kW larger than that of mismatch-only feedback, which updates each ceiling from the participant's delivery-to-award ratio without checking whether reallocation would increase export elsewhere.

The two baselines can be reconciled directly. Relative to forecast-based requests, mean delivery changes are -21.7 kW without feedback, -6.5 kW with mismatch-only feedback and $+3.2$ kW with ECF. Thus the 24.9-kW feedback gain removes the 21.7-kW loss and leaves the 3.2-kW increase. Table S9 gives the unrounded values.

ECF retains 79.4% of the larger-request group's delivery gain obtained without feedback over forecast-based requests. Total delivery, initially below that reference without feedback, rises above it with ECF. Supplementary section S3.5 reports the group decomposition and the numerical comparisons with each baseline. At $\vartheta = 10$, ECF raises mean delivery but increases the economic externality share from zero to 1.3%. A higher mean delivery can therefore coexist with more intervals classified as request–access–delivery externalities.

To isolate the effect of where requests are reduced, the equal-total control removes the same total request as ECF in each interval but distributes the reduction proportionally across connections. ECF exceeds this control in 95.6% of the 90 matched trajectories at zero shortfall cost. Thus the mean advantage comes from where requests are reduced, rather than their total reduction. Supplementary section S3.5 reports comparisons at the other cost settings.

4.2.3. Update strength and generation recovery

Selecting a useful interval is insufficient if the next ceiling is too tight. At zero shortfall cost, replacing the ceiling with one interval's target, rather than using a gain of 0.5, changes the 3.2-kW increase over forecast-based requests into a 23.4-kW decrease (Figure 4). This unit-gain response can restrict export when output subsequently recovers. The intermediate gain of 0.5 gives the largest mean delivery among the three tested gains; the reversal also occurs at the other cost settings.

Almost all controlled replays identify an alternative allocation with higher total delivery (supplementary section S3.5). These trajectories therefore mainly test where to reduce requests and how strongly to update ceilings. The public-data comparison below tests the choice of update intervals when such opportunities are less frequent.

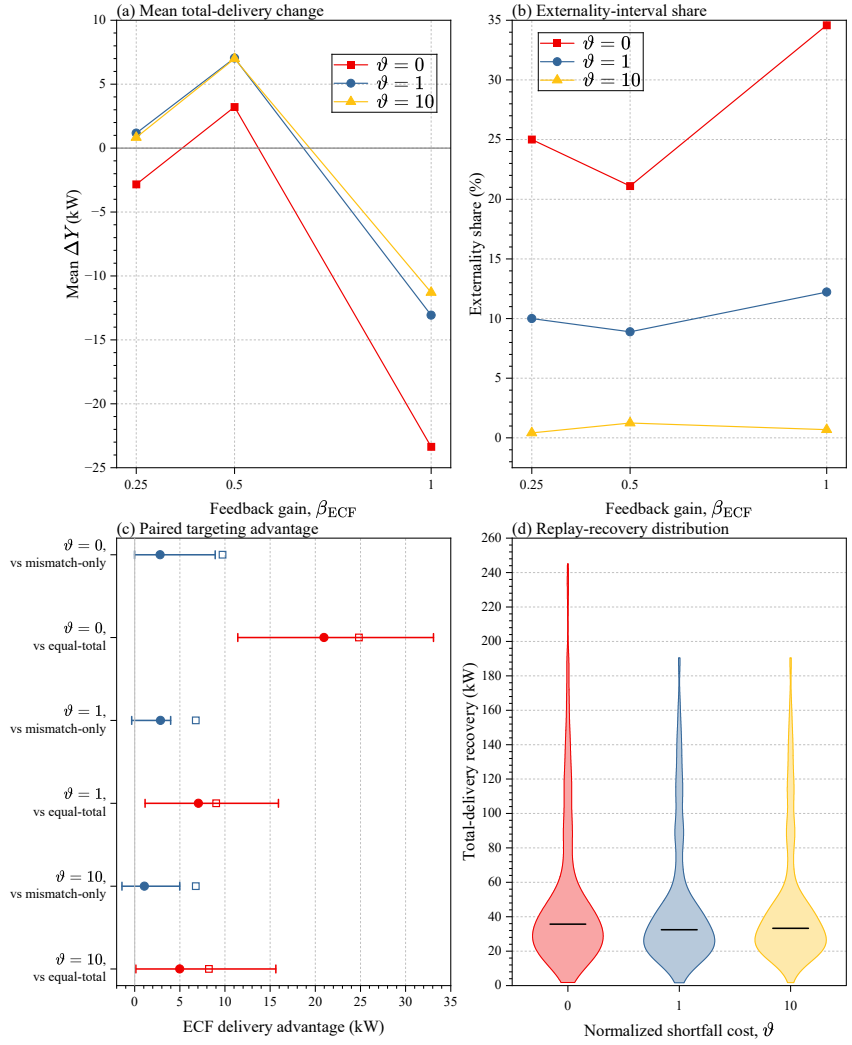


Figure 4: Targeted feedback preserves delivery, but a unit-gain update reverses the improvement in the controlled cases. (a) Mean signed change in total delivery relative to forecast-based reference requests and (b) externality-interval share for the three tested feedback gains. The series denote $\vartheta = 0, 1, 10$. (c) ECF delivery advantage in matched trajectories. Rows labeled "vs mismatch-only" compare ECF with mismatch-only feedback. Rows labeled "vs equal-total" compare ECF with the equal-total control. Panels (c) and (d) use the primary gain, $\beta_{\mathrm{ECF}} = 0.5$. Filled circles show medians, open squares show means, and whiskers show interquartile ranges. The gray line marks zero. (d) Distribution of total delivery recovered when the replay meets all four conditions. Violin width shows empirical density, and the black horizontal segment shows the median.

4.3. Output timing and allocation rules alter the delivery gain

The delivery advantage persists across the five allocation rules, but its size depends on how each rule distributes access. At $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$, ECF exceeds equal-total reduction by 9.7–50.8 kW on average across the rules (Figure 5). Mean delivery also exceeds no feedback under all five rules.

The three temporal profiles preserve each participant's available-export values and frequencies. They change their order and, because participants have different phase offsets, which output states occur together across connections. Clustered low output gives the largest mean gain, intermittent low output the smallest, and the shock–recovery sequence lies between them. Thus the order of output changes and their coincidence across connections affect the feedback benefit, even when each participant's distribution of output values is unchanged.

The advantage also remains positive for one changed requester, five nearby requesters and five dispersed requesters. The five-participant sets differ in identity and location, together with their associated capacities and requests. Each group's composition is fixed within its paired comparisons. The two reactive-power settings give nearly identical mean gains. Supplementary section S3.5 collects the factor-level values and the allocation-equality results alongside Figure S3.

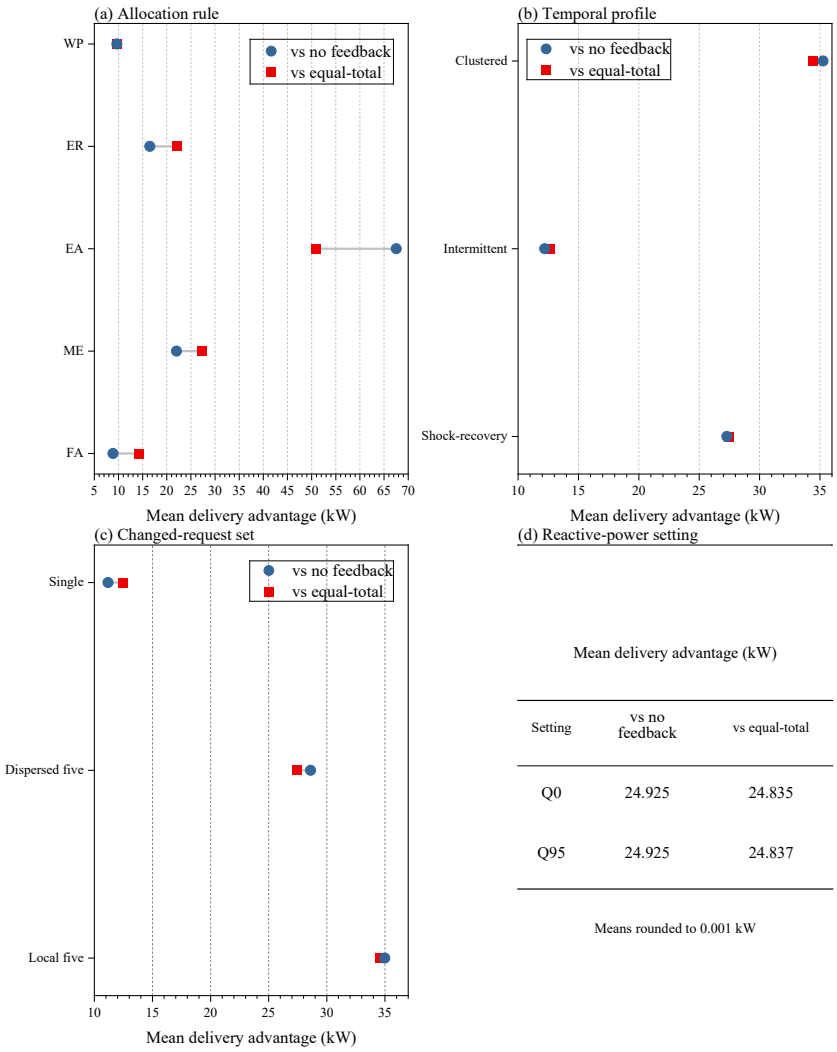


Figure 5: The controlled delivery advantage persists across allocation rules and output timing at $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$. Markers are factor-level means, grouped by (a) allocation rule, (b) temporal profile and (c) changed-request set. Panel (d) tabulates means for the two reactive-power settings to show their close agreement. Blue circles compare ECF with no feedback. Red squares compare ECF with the equal-total control. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively. Q0 sets participant reactive power to zero, whereas Q95 uses 0.95-power-factor absorption. The temporal profiles preserve individual output frequencies but change ordering and cross-connection coincidence.

4.3.1. Allocation-domain use, AC feasibility and equality

In the controlled dynamic trajectories, 25,918 of 25,920 allocations lie within 5×10^{-6} of the prescribed normalized-radius limit of 3.5. The calibrated lower-voltage and branch-loading inequalities retain minimum residuals of 0.003894 pu and 0.004550 MVA, respectively, before the small numerical buffers specified in Supplementary section S8. The dynamic comparisons therefore measure reallocation within this nearly saturated prescribed domain. Supplementary section S8 defines the residuals and covers all dynamic controls and parameter combinations.

AC power flow converges for the controlled allocation and delivery states, with active- and reactive-power balance, bus voltages and apparent power at both branch ends satisfying the specified limits and synthetic branch ratings. Figure S3 and Section S3.5 report the loading and voltage margins.

More delivery need not mean more equal access. The capacity-normalized Jain index in Figure S3 measures equality of awards, not social fairness. At $\vartheta = 0$, the mean index is higher with ECF than with no feedback and equal-total reduction under all five rules. Four rules nevertheless remain below the forecast-based reference.

4.4. Public-data simulations show when targeting raises export

The controlled trajectories mainly test whom to target and how strongly to update. The public-data comparisons below focus on when a shortfall should trigger an update. In this simulation, ECF increases total export by changing both whose requests are reduced and when reductions occur. The gains vary across days and participants. The matched controls test whether selecting update intervals adds value beyond updating less often.

4.4.1. Reallocation increases delivered energy

Across 80 Australian network-days, ECF delivers 239.36 MWh of modeled export: 1.39 MWh more than either no feedback or equal-total request reduction (Figure 6). The primary comparison uses weighted proportional allocation on the unbalanced network and household output histories described in Section 3.3. Relative to no feedback, the increase is 0.5851%.

The increase has two components. Relative to no feedback, ECF authorizes an additional 1.206 MWh and leaves 0.186 MWh less unused, giving 1.392 MWh more delivery through (7). Equal-total reduction has almost no effect here: proportional request cuts preserve the relative requests, and the allocation algorithm can offset the cuts by increasing its common factor while the network limit remains binding. Targeted feedback changes those relative requests and hence the distribution of awards. The scaling condition holds in 967 of the 969 half-hours with a uniform request reduction (supplementary section S4). This explains why equal-total reduction nearly reproduces no feedback for this rule; it is a control for the placement of reductions, not an optimized alternative dispatch. Section S11.1 gives the full energy balance.

The gain varies across days: 44 improve, 32 lose delivery and four remain unchanged relative to either control. Against no feedback, the mean is 17.4 kWh per day, with a dependence-adjusted 95% interval of 5.1–29.7 kWh. A continuous 35-day block contributes 1.06 MWh of the 1.39-MWh gain; the remaining dates give a positive total, but the interval for their mean daily difference includes zero (supplementary section S10.3).

The extra export at the connections reduces net upstream import by 1.38 MWh relative to either control, retaining about 98.9% of the export gain at the feeder source. The feeder remains a net importer; supplementary section S11.1 gives the circuit-loss and passive-element terms.

Previous-day persistence illustrates why utilization alone is insufficient. It uses 89.0% of its authorized energy but delivers 50.33 MWh less than ECF. Without the forecast increment, its requests are too low in many high-output intervals. The higher utilization therefore accompanies less export.

The public-study trajectories satisfy the stated allocation, customer-voltage and transformer-loading criteria (supplementary section S14.3).

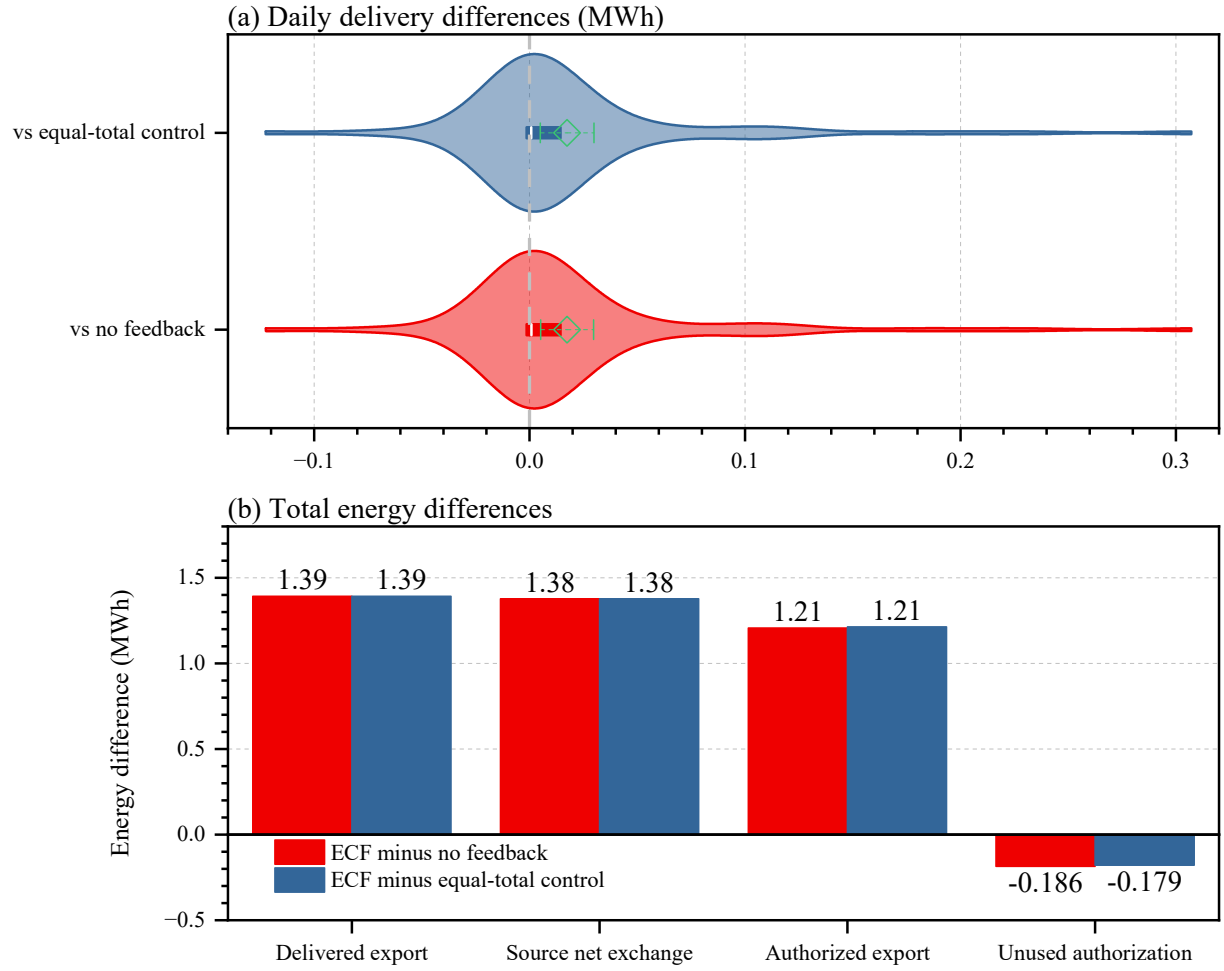


Figure 6: Modeled delivery gains and their energy balance in the Australian public-data simulation. Available export is constructed from PV and load profiles. (a) Paired daily delivery differences for ECF against no feedback and the equal-total control. Violin width shows empirical density. Dark boxes span the 25th–75th percentiles, and white ticks show medians. Green diamonds show means; capped green lines show 95% intervals for the mean daily difference, adjusted for short-range dependence between neighboring dates. (b) Aggregate 80-day differences in delivered export, source-terminal net exchange, authorization, and unused authorization. Each difference is ECF minus the indicated control. Delivery equals authorization minus its unused portion. Here the delivery gain accompanies increased authorization and reduced unused authorization. Positive source-terminal differences indicate reduced net upstream import over the assessed period.

4.4.2. Selecting update intervals improves delivery

ECF lowers ceilings toward completed delivery in 12.6% of half-hours. Previously reduced ceilings can remain below their connection limits while recovering between these updates. To test the benefit of this selection, we first remove the replay test but retain the ceiling formula and gain. Updating after every unused award delivers 6.42 MWh less than ECF, with ECF ahead on 75 of 80 days. Although this control reduces unused allocations further, it also reduces delivered export (Table 2).

The first control also updates much more often. The second keeps the same target and gain but randomly selects unused-award intervals at approximately ECF's frequency (Section 3.1). ECF delivers more than every one of the twelve schedules and 1.34 MWh more than their mean, despite leaving almost the same total authorization unused. Its mean paired daily advantage is 16.8 kWh, with a dependence-adjusted 95% interval of 7.8–25.8 kWh.

Thus updating less often does not by itself reproduce ECF's gain. Replay-based selection preserves more capacity for subsequent export than random selection at a similar frequency. Restriction sizes and durations still vary with each schedule's allocation and ceiling history. Supplementary sections S11.2–S11.3 give the control design and full results.

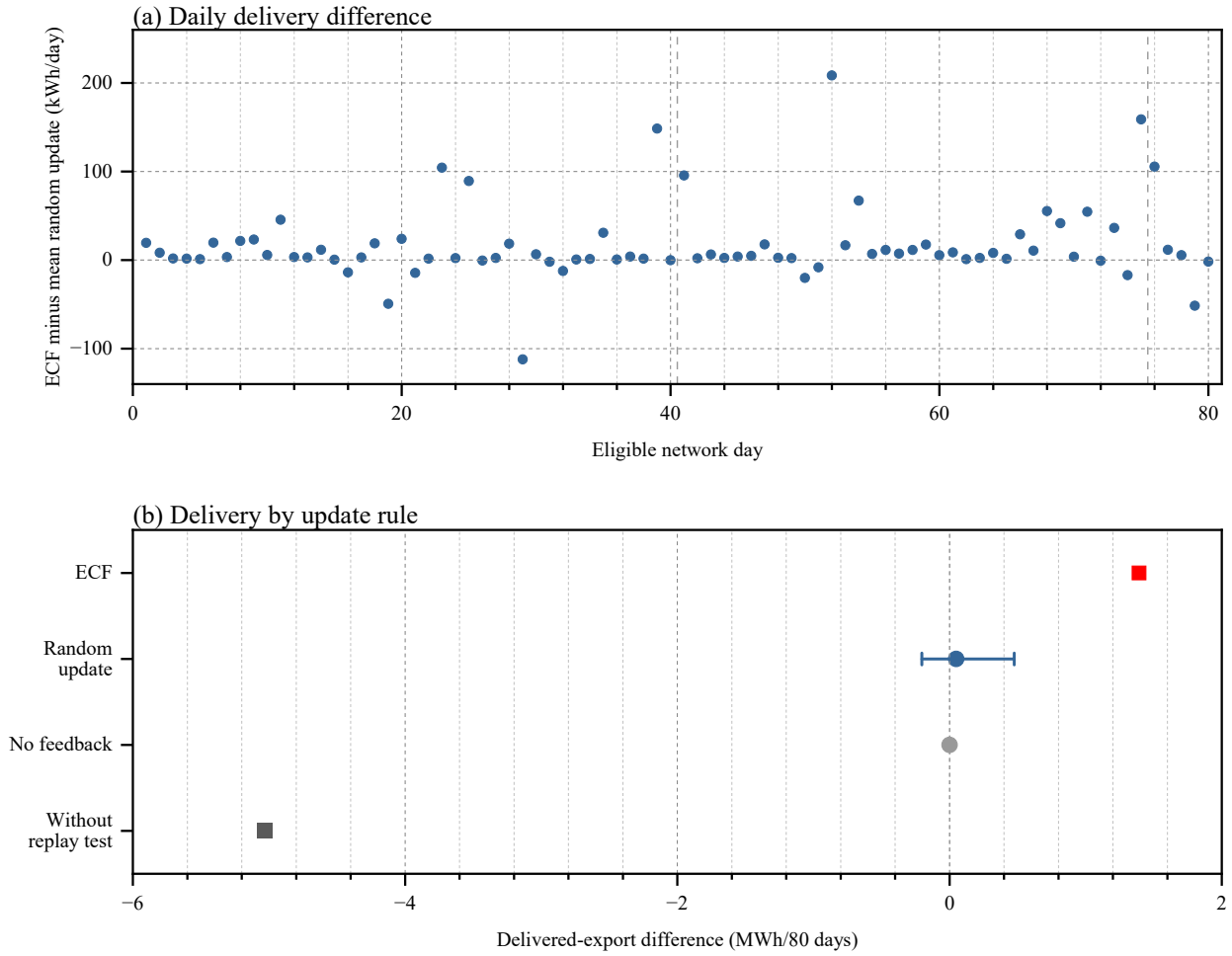


Figure 7: Replay-selected updates recover more modeled delivery than comparably frequent random updates in the Australian simulation. (a) Paired daily modeled-delivery difference between ECF and the mean of 12 frequency-calibrated random updates. The vertical dashed lines bound the longest continuous 35-day block. (b) Eighty-day delivery relative to no feedback. The blue marker is the mean across 12 fixed random schedules; its horizontal range spans their individual totals. The red square marks ECF; gray markers denote no feedback and updating after every unused award. The number of half-hours selected for a delivery-target update is 483 for ECF, 463–519 for the random schedules, 0 for no feedback and 3,086 for updating after every unused award. Ceiling recovery still operates in other intervals. The schedules are retrospectively calibrated to approximately ECF's update frequency, without using replay to choose their intervals. Their independent ceiling histories need not give identical update magnitudes or durations.

Table 2
Modeled export over the same 80 Australian network days. The update without a replay test uses ECF's ceiling formula and gain after every unused award. The frequency-calibrated random row averages 12 replay-free schedules with the same formula and gain. The equal-total control matches ECF's request reduction in each half-hour. All energy totals are in MWh and are rounded independently.

Control	Authorized	Unused	Delivered
No feedback	328.490	90.526	237.963
Equal-total reduction	328.483	90.519	237.964
Update without replay test	305.159	72.226	232.933
Frequency-calibrated random update	328.357	90.345	238.012
ECF	329.696	90.341	239.356

Figure 7 shows the daily differences and the spread of total delivery across the random schedules.

4.4.3. Update gain and record errors affect delivery

Full replacement of the ceiling by the latest target, or unit gain, loses 3.04 MWh relative to no feedback (Figure 8). The loss is concentrated in recovering generation. Relative to gain 0.5, unit gain loses a net 5.47 MWh in connection–intervals where its previous update tightened the ceiling, available output then rose, and its ceiling was below both the raw request and available export. Gains of 1.04 MWh over the remaining connection–intervals only partly offset this loss. Supplementary section S12.1 defines this partition of the two feedback trajectories. Retaining ceilings across consecutive days preserves both the moderate-gain advantage and the unit-gain loss (supplementary sections S12.1–S12.2).

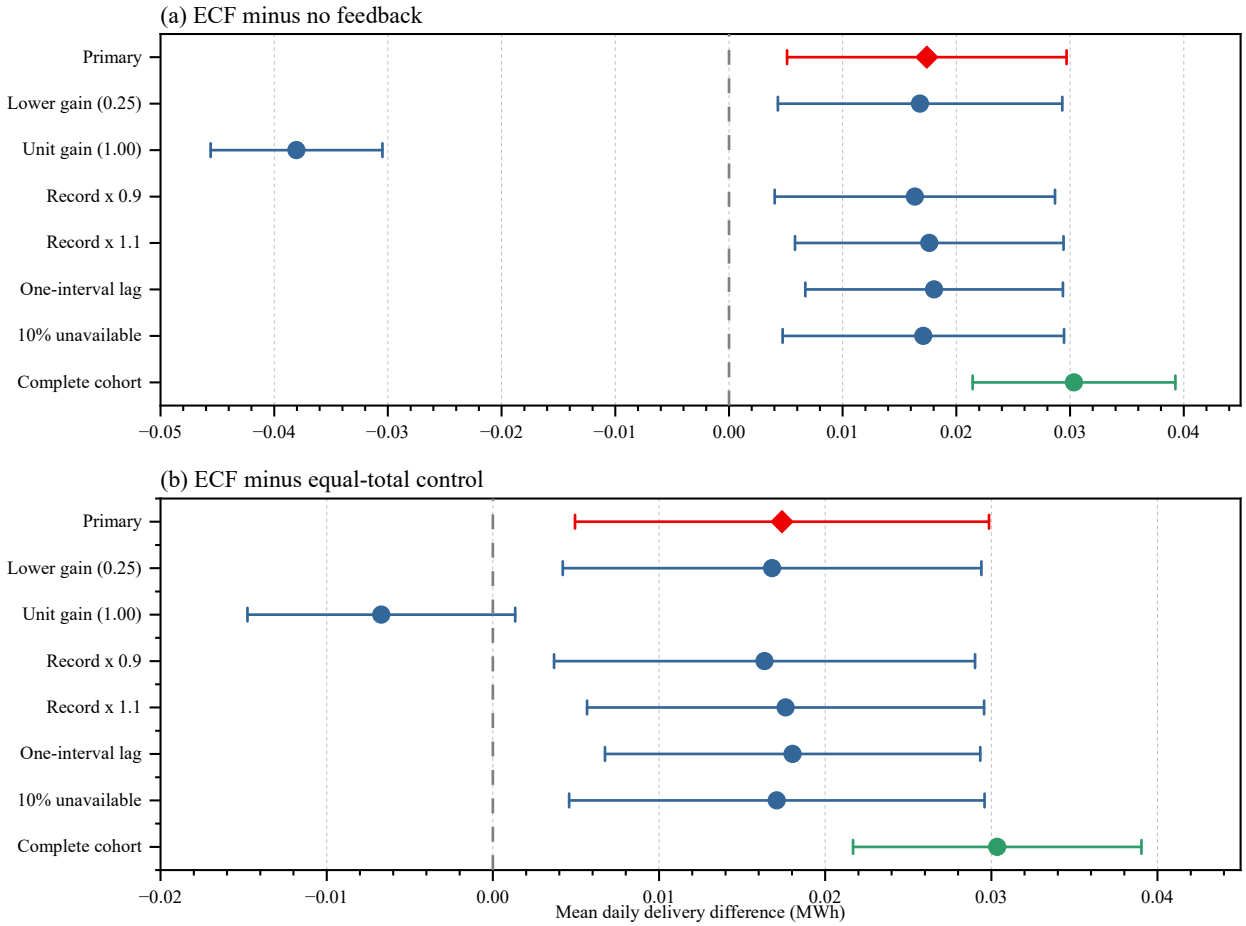


Figure 8: Gain, record quality and customer coverage change the Australian delivery result. Markers show paired mean delivery differences per network day. Bars show 95% intervals adjusted for short-range dependence between neighboring dates. (a) ECF minus no feedback. (b) ECF minus the equal-total control. The red diamond denotes the primary unperturbed-proxy result, and blue circles denote gain or record changes. Green markers denote the complete-customer cohort, which uses the same load multiplier but changes customer composition and calendar coverage (361 days rather than 80). Each transformed available-export record is bounded below by modeled delivery. The zero line separates positive from negative mean delivery differences.

Record quality changes which intervals trigger an update. Scaling or delaying the available-export record, or filling missing records with completed delivery, leaves a positive mean advantage over both no feedback and equal-total reduction in the reported tests. Here the altered records change the estimated delivery gain, while AC checks use unperturbed profile-derived availability (supplementary section S12.1). These tests vary a simulated input around known profile-derived availability; they do not estimate the error of a field instrument or availability estimator.

The error-bound comparison also uses the perturbed record in the replay power flow. Both controllers apply this AC check. One calculates the delivery gain directly from the record; the other first subtracts a prescribed error bound, retaining completed delivery as a minimum. This lower-bound controller avoids

updates that fail the recovery test under unperturbed availability, but misses some useful updates when the record understates output.

At the largest negative bias, the lower-bound controller delivers 269 kWh less than the controller using the record directly; both still exceed no feedback. For the prescribed positive bias, subtracting the bound exactly cancels the imposed error and restores unperturbed availability. Supplementary section S12.3 gives the error model and the independent delivery-recovery and AC checks against unperturbed availability.

4.4.4. Location and PV capacity change the benefit

Across five additional placements of the same household profiles, the gain ranges from about 0.9 to 5.1 MWh over no feedback and equal-total reduction. Each placement has a positive pointwise uncertainty interval for its mean daily difference. Varying PV capacity and connection bounds at fixed locations also preserves a positive gain, but more installed PV does not consistently produce a larger gain. The gain per MW of installed PV falls from 0.81 to 0.27 MWh/MW over the same 80 evaluation days across the tested capacity range. The benefit therefore depends on electrical placement as well as installed capacity. Supplementary sections S13.1–S13.2 give the matched settings and full results.

The network comparisons include gains, a loss and unchanged outcomes. In 13 LV sections of the parent feeder, ECF exceeds equal-total reduction in every section and no feedback in 12; LV26 loses 3.7 kWh against no feedback. Of nine representative published LV models, six gain delivery, including one near-zero gain. The remaining three never meet the replay conditions and are unchanged (Figure S5). Supplementary sections S13.3–S13.4 give the network comparisons under the common scenario construction.

Using the complete-customer cohort extends the comparison to 361 days at the same load scale. ECF gains 30.4 kWh per day over no feedback and equal-total reduction, with positive uncertainty intervals for both mean differences. Both customer composition and calendar coverage change in this extension (supplementary sections S10.3 and S12.1).

4.4.5. Individual outcomes and request adjustment

The aggregate gain includes both individual gains and losses. Among 418 participants in the primary feeder, 330 deliver more than without feedback over the 80 days and 88 deliver less; the largest individual loss is 7.3 kWh. All 418 participants experience at least one day with lower delivery, and the longest run of such days is five. A positive aggregate result therefore does not imply that every participant or day benefits (supplementary section S14.1).

The preceding comparisons use a fixed forecasting rule for requests. When participants instead change the probabilities of choosing different request levels using their own past deliveries, ECF still gains 1.44 MWh over no feedback and 4.53 MWh over the same ceiling update without a replay test. Each control produces its own delivery history and therefore its own request adjustments. These differences use the adaptation experiment's own no-feedback and no-replay baselines, not the primary energy totals in Table 2 (supplementary section S14.2).

5. Discussion

Using delivery history to allocate export capacity requires two decisions: whether a shortfall warrants a later restriction, and how much restriction to carry forward. ECF makes the first decision using the delivery achievable under a feasible redistribution in the completed interval. Its gradual update and recovery then regulate the effect on future access. The public comparisons show a gain over both indiscriminate and comparably frequent random updates. The unit-gain reversal shows why selecting useful past opportunities must be assessed together with the export lost during recovery.

This distinction also changes the performance measure. Updating after every unused award leaves less authorization unused than ECF but delivers less electricity. Conversely, ECF's public-study gain is mainly associated with more authorized export, with a smaller contribution from reduced unused authorization. Among the tested controls, minimizing the unused amount would therefore select a control with lower delivered export. The relevant balance is additional export elsewhere minus export lost under subsequent limits.

5.1. Implications for flexible export operation

For an operator retaining a request-based allocation rule, completed delivery can inform the next request limit without changing that rule's objective. The comparison should follow delivered energy through low-output and recovery periods, alongside participant losses and restriction durations. Shortfall settlement and forecasting remain distinct operating choices: they change incentives or expected inputs before allocation, whereas ECF checks a possible redistribution after delivery. Faster reassignment within an interval is another option where the operating arrangements permit it.

The proposed test requires additional information and computation. In the public study, the mean power-flow workload is 21.1 calls per half-hour with ECF versus 11.4 without feedback, because replay is attempted even when no update follows. This workload accompanies a 0.59% delivered-energy gain, almost all of which appears as reduced upstream import. An operational assessment must relate that gain to the time and cost of obtaining valid availability records and performing the calculations. The present call counts quantify computational work, not execution time or economic payback.

The matched controls identify the roles of update selection and strength within the retained allocation rule. The public random-update schedules approximately match update frequency; they do not match each reduction's magnitude and duration. Forecast-aware multi-period dispatch and history-weighted fairness optimization address broader scheduling or allocation choices; some also change network configuration [24]. Comparing those operating strategies with ECF would require common information and network constraints.

5.2. Individual outcomes and information requirements

ECF tests a joint request reduction for the participants with unused awards. It does not determine how much each member contributes to the additional export, so it cannot by itself assign an individual penalty. The participant losses reported here also warrant attention when assessing group updates. ECF's mean advantage persists when participants adjust requests using their own delivery histories. That experiment represents adjustment based on past outcomes, rather than strategic planning of requests in anticipation of future ceilings.

The controlled reallocation results use the prescribed allocation domain examined in Section 4.3.1 and supplementary section S8. The public study uses published network structures and measured household histories, with constructed requests and available export. Its checks cover customer voltage and native transformer ratings, not unavailable line ampacities. These checks define the physical scope of the public-data results.

The information requirement is equally important. Replay needs linked requests, awards, delivery and available-export estimates for the same connections and intervals. Metered delivery alone cannot reveal output withheld by an export limit. Storage, inverter control or flexible demand must enter the replay model if a different award changes their available export. A validated error bound can prevent an overestimate of recoverable delivery, at the cost of missing some useful updates. The specified proxy perturbations test that trade-off but do not validate an operational estimator.

6. Conclusions

ECF uses the delivery achievable under a feasible alternative allocation to decide whether a past shortfall warrants lower future request limits. Gradual updates and recovery control the resulting restriction on later generation while retaining the existing allocation rule.

The controlled comparisons show the value of targeting request reductions within the prescribed allocation domain. The public-data study further shows the value of selecting update intervals: ECF delivers 1.39 MWh more modeled export than no feedback and 1.34 MWh more than the mean random-update comparison. Direct replacement of ceilings by low-output targets reverses the benefit. These findings support selective, gradual updates evaluated through generation recovery. Among these controls, the largest reduction in unused awards would favor a control that delivers less electricity.

For request-based export operation, the relevant outcome is electricity delivered over successive intervals, including export enabled elsewhere and generation restricted during recovery. Participant losses and the cost of obtaining available-output information also enter an operational assessment. Such assessments can guide the use of existing network capacity alongside forecasting, settlement and allocation.

CRediT authorship contribution statement

Junjie Huang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Data curation, Writing – original draft. **Tao Yu:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing. **Zhenning Pan:** Methodology, Validation, Writing – review & editing. **Yufeng Wu:** Methodology, Validation, Writing – review & editing. **Wencong Xiao:** Methodology, Validation, Writing – review & editing. **Rankai Zhu:** Validation, Data curation, Writing – review & editing. **Jie Huang:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

Figure source data accompany this article in Supplementary data S1. Calculated results, simulation and analysis code, parameter files and reproduction instructions are available at `https://github.com/MarxHuang/ecf-export-access`. The repository release note identifies the manuscript and supplementary files corresponding to this revision. Questions can be directed to the corresponding author, Zhenning Pan (scutpanzn@163.com). The original 141-bus network, CSIRO/GridQube network collection (doi:10.25919/ghnz-bk28) and Ausgrid Solar Home Electricity Data remain with their original distributors under their respective terms. The repository records source identifiers and preprocessing steps; raw third-party records are not redistributed.

Supplementary material

Supplementary information contains the derivations, model specifications, Figures S1–S6 and additional comparisons. Supplementary data S1 contains eight figure source-data workbooks and a figure-to-worksheet map.

Acknowledgements

This work was supported by State Grid Electric Power Research Institute Co., Ltd., under the project “Interaction and Coordination Technology of Cyber-Physical-Social Elements (II)” [grant number SGNR0000KJJS2505736].

References

[1] R. Wang, H. Ji, P. Li, H. Yu, J. Zhao, L. Zhao, Y. Zhou, J. Wu, L. Bai, J. Yan, C. Wang, Multi-resource dynamic coordinated planning of flexible distribution network, Nature Communications 15 (2024) 4576. doi:10.1038/s41467-024-48862-5.

[2] National Energy Administration of China, Policy interpretation of the measures for the development and construction of distributed photovoltaic power generation, in Chinese; accessed 3 September 2026 (Jan. 2025). URL https://www.nea.gov.cn/20250123/d38e5436b4d04159863ddbc10a6ede10/c.html

[3] SA Power Networks, Distribution annual planning report 2024/25 to 2028/29, Tech. rep., SA Power Networks, accessed 3 September 2026 (Dec. 2024). URL https://www.sapowernetworks.com.au/public/download.jsp?id=9716

[4] California Public Utilities Commission, Resolution e-5230: Proposals to implement limited generation profiles and time-varying inverter export values, Tech. rep., California Public Utilities Commission, issued 9 December 2022; accessed 3 September 2026 (Dec. 2022). URL https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M499/K779/499779492.PDF

[5] European Parliament and Council of the European Union, Directive (EU) 2024/1711 amending directives (EU) 2018/2001 and (EU) 2019/944 as regards improving the union’s electricity market design, Official Journal of the European Union, accessed 3 September 2026 (2024). URL https://eur-lex.europa.eu/eli/dir/2024/1711/oj

[6] K. Petrou, A. T. Procopiou, L. Gutiérrez-Lagos, M. Z. Liu, L. F. Ochoa, T. Langstaff, J. M. Theunissen, Ensuring distribution network integrity using dynamic operating limits for prosumers, IEEE Trans. Smart Grid 12 (5) (2021) 3877–3888. doi:10.1109/TSG.2021.3081371.

[7] M. Z. Liu, L. F. Ochoa, P. K. C. Wong, J. M. Theunissen, Using OPF-based operating envelopes to facilitate residential DER services, IEEE Trans. Smart Grid 13 (6) (2022) 4494–4504. doi:10.1109/TSG.2022.3188927.

[8] M. M. Hoque, M. Khorasany, M. I. Azim, R. Razzaghi, M. Jalili, Dynamic operating envelope-based local energy market for prosumers with electric vehicles, IEEE Trans. Smart Grid 15 (2) (2024) 1712–1724. doi:10.1109/TSG.2023.3302270.

[9] P. Scott, J. Iria, D. Gordon, A. Fraser, B. Weise, A. Attarha, S. M. Noori R. A., Shaped operating envelopes: Technical design and implementation report, Tech. rep., The Australian National University, project Converge; accessed 24 September 2026 (Feb. 2023).
URL https://arena.gov.au/assets/2023/02/shaped-operating-envelopes-technical-design-implementation-report.pdf

[10] M. Andoni, V. Robu, W.-G. Früh, D. Flynn, Game-theoretic modeling of curtailment rules and network investments with distributed generation, Applied Energy 201 (2017) 174–187. doi:10.1016/j.apenergy.2017.05.035.

[11] C. Gorrasi, K. Bruninx, E. Delarue, Comparison of market designs ensuring network integrity in low voltage distribution systems with high DER penetration, Applied Energy 372 (2024) 123804. doi:10.1016/j.apenergy.2024.123804.

[12] B. Liu, J. H. Braslavsky, Robust dynamic operating envelopes for DER integration in unbalanced distribution networks, IEEE Trans. Power Syst. 39 (2) (2024) 3921–3936. doi:10.1109/TPWRS.2023.3308104.

[13] B. Liu, J. H. Braslavsky, N. Mahdavi, Linear OPF-based robust dynamic operating envelopes with uncertainties in unbalanced distribution networks, J. Mod. Power Syst. Clean Energy 12 (4) (2024) 1320–1326. doi:10.35833/MPCE.2023.000653.

[14] A. Attarha, S. M. Noori R. A., P. Scott, S. Thiébaux, Network-secure envelopes enabling reliable DER bidding in energy and reserve markets, IEEE Trans. Smart Grid 13 (3) (2022) 2050–2062. doi:10.1109/TSG.2021.3138099.

[15] K. Petrou, M. Z. Liu, A. T. Procopiou, L. F. Ochoa, J. M. Theunissen, J. Harding, Operating envelopes for prosumers in LV networks: A weighted proportional fairness approach, in: Proc. IEEE PES Innov. Smart Grid Technol. Europe (ISGT-Europe), 2020, pp. 579–583. doi:10.1109/ISGT-Europe47291.2020.9248975.

[16] M. R. Alam, P. T. H. Nguyen, L. Naranpanawe, T. K. Saha, G. Lankeshwara, Allocation of dynamic operating envelopes in distribution networks: Technical and equitable perspectives, IEEE Trans. Sustain. Energy 15 (1) (2024) 173–186. doi:10.1109/TSTE.2023.3275082.

[17] J. J. Cuenca, B. P. Hayes, Non-bias allocation of export capacity for distribution network planning with high distributed energy resource integration, IEEE Trans. Power Syst. 37 (4) (2022) 3026–3035. doi:10.1109/TPWRS.2021.3124999.

[18] Z. Jiang, Y. Guo, Bargaining-based approach for dynamic operating envelope allocation in distribution networks, IEEE Trans. Smart Grid 16 (5) (2025) 3588–3600. doi:10.1109/TSG.2025.3566419.

[19] Y. Chen, C. Zhao, S. H. Low, A. Wierman, An energy sharing mechanism considering network constraints and market power limitation, IEEE Trans. Smart Grid 14 (2) (2023) 1027–1041. doi:10.1109/TSG.2022.3198721.

[20] G. Tsaousoglou, J. S. Giraldo, P. Pinson, N. G. Paterakis, Mechanism design for fair and efficient DSO flexibility markets, IEEE Trans. Smart Grid 12 (3) (2021) 2249–2260. doi:10.1109/TSG.2020.3048738.

[21] G. Belgioioso, W. Ananduta, S. Grammatico, C. Ocampo-Martinez, Operationally-safe peer-to-peer energy trading in distribution grids: A game-theoretic market-clearing mechanism, IEEE Trans. Smart Grid 13 (4) (2022) 2897–2907. doi:10.1109/TSG.2022.3158442.

[22] University of Melbourne, Department of Electrical and Electronic Engineering, Project EDGE: Determining the impact of update frequency on operating envelope efficacy, Tech. rep., Australian Energy Market Operator, prepared for Project EDGE (March 2023).
URL https://www.aemo.com.au/-/media/files/initiatives/der/2023/project-edge-determining-the-impact-of-update-frequency-on-operating-envelope-efficacy--workstream-3.pdf

[23] H. Moring, X. Farrell, J. L. Mathieu, Fair-over-time distributed energy resource coordination, in: Proc. 60th Annu. Allerton Conf. Commun., Control, Comput., 2024, pp. 1–8. doi:10.1109/Allerton63246.2024.10735316.

[24] R. K. Gupta, D. K. Molzahn, Improving fairness in photovoltaic curtailment via feedback-driven daily topology reconfiguration in power distribution networks, Applied Energy 400 (2025) 126543. doi:10.1016/j.apenergy.2025.126543.

[25] S. Sweeney, P. Kilby, B. Penney, K. Moghaddasi, S. Mudiyanselage, Stateful pricing and allocation for repeated constrained DER coordination in distribution networks, arXiv:2606.22463, preprint, version 1 (2026). doi:10.48550/arXiv.2606.22463.

[26] M. Vassallo, A. Bolland, A. Bahmanyar, L. Wehenkel, L. Duchesne, D. Liu, S. Khaskheli, A. H. Thuc, P. P. Vergara, A. Anvari-Moghaddam, S. Gerard, D. Ernst, SecuLEx: A secure limit exchange market for dynamic operating envelopes, Electr. Power Syst. Res. 263 (2027) 113736. doi:10.1016/j.epsr.2026.113736.

[27] X. Yu, Y. Xue, Smart grids: A cyber–physical systems perspective, Proc. IEEE 104 (5) (2016) 1058–1070. doi:10.1109/JPROC.2015.2503119.

[28] Y. Xue, X. Yu, Beyond smart grid—cyber–physical–social system in energy future [point of view], Proc. IEEE 105 (12) (2017) 2290–2292. doi:10.1109/JPROC.2017.2768698.

[29] H. M. Khodr, F. G. Olsina, P. M. De Oliveira-De Jesus, J. M. Yusta, Maximum savings approach for location and sizing of capacitors in distribution systems, Electr. Power Syst. Res. 78 (7) (2008) 1192–1203. doi:10.1016/j.epsr.2007.10.002.

[30] F. Geth, R. Heidari, J. Clark, K. Lucas, N. Nimalsiri, Realistic australian medium voltage feeder with associated low voltage feeders (2025). doi:10.25919/ghnz-bk28.
URL https://doi.org/10.25919/ghnz-bk28

[31] Ausgrid, Solar home electricity data, 1 july 2010 to 30 june 2013, 300 gross-metered homes; accessed 13 August 2026 (2014).
URL https://data.gov.au/data/dataset/nsw-solar-home-electricty-data

[32] Clean Energy Regulator, Quarterly carbon market report: March quarter 2025, Tech. rep., Australian Government Clean Energy Regulator, accessed 13 August 2026 (2025).
URL https://cer.gov.au/document/quarterly-carbon-market-report-march-quarter-2025